2026/10/13

Bassam's Unified Quantum Theory

Bassam Muwafaq Salman

TEACHER

General Directorate of personnel (*Civilian Personnel*), *Training and Education Development Section, Iraqi Ministry of Defense*.

E- mail: basam.muwafaq86@gmail.com. basam.classic86@gmail.com

ORCID: 0009-0009-0659-7227

An earlier version of this work is available on Zenodo (DOI: 10.5281/zenodo.23328953) and on HAL (hal-05796325).

The priority date for the fundamental equations of Bassam's Unified Quantum Theory, in the field of Theoretical High Energy Physics (hep-th), authored by Bassam Muwaffaq Salman Abdul Karim, is established as 13/10/2026.

This specific preprint was published on Figshare on 15/10/2026 (DOI: 10.6084/m9.figshare.34532949).

Abstract

This work presents the first publication of a novel theory, termed Bassam's Unified Quantum Theory. The theory postulates that the characteristic time of any physical system can be derived directly from Bassam's fundamental time equation, $t_B = S_{VN}/\pi$, (in natural units) $\hbar = k_B = c = 1$, where time emerges from S_{VN}, which represents the quantum information entropy (von Neumann entropy) and is a dimensionless quantity, here π is a mathematical constant. Applying the equation to the electron in the maximally mixed state, where the von Neumann entropy (unpolarized spin) is $S_{VN}^{(e)} = \ln 2$, yields a theoretical characteristic time for the electron of $t_B^e = 2.846 \times 10^{-22}$s, upon restoring SI units. This time is interpreted as a fundamental timescale of the system, representing the quantum time of the electron. The theory further yields experimentally testable predictions through the solution of the Dirac-Bassam equation, a theoretical extension of the classical Dirac equation. Foremost among the theory's predictions is the existence of new excited particles-namely, excited electrons (the Bassam electron) with a predicted mass of $m_B^{(e)} \approx 15.1\ \mathrm{MeV/c^2}$. The particle is experimentally detectable in particle colliders, particularly in high-energy physics experiments such as **Belle II** or **LHCb**, where it can be produced via the decay channel $e_B^- \rightarrow e^- + \gamma$. The theory provides a unified framework that connects quantum mechanics, relativity, and quantum information theory.

Keywords: Quantum Information, Excited Bassam Electron, Emergent Time.

1. INTRODUCTION

In modern theoretical physics, the question of the fundamental nature of time persists as one of the most difficult questions to answer and one of the profound mysteries of nature [1]. In this theoretical context, we present a new contribution to understanding how time emerges revealing that the same mechanism that leads to the emergence of space can be incorporated within a unified framework. Specifically, time emerges through the entropy of quantum information [2,3]. This perspective finds its foundation in the mathematical regularity that bridges physical and mathematical relations across different levels. Indeed, a common feature underlies all equations in modern theoretical physics From Einstein's equation in general relativity and Schrödinger's equation for the energy and momentum of the atomic particle [4]. to Dirac's equation for the electron in quantum physics [5]. this common feature is that they all preserve quantum information [6]. and all of these equations preserve conserved physical quantities [7,8]. Consequently we obtain a coherent and concise mathematical framework that establishes a unified quantum connection among these theories through the introduction of what we term Bassam's Unified Quantum Theory. Central to this theory is a fundamental relation between the time scale t_B and the quantum information entropy (von Neumann entropy), S_{VN}, which is a dimensionless quantity inherent to the system. which is given by Bassam's fundamental time equation, $t_B = (S_{VN}/\pi) \cdot t_0$. (upon restoring SI units), where t_0 represents the natural time scale of the system. By applying this equation to an electron in its maximally mixed state, $S_{VN}^{(e)} = \ln 2$, we obtain the Bassam quantum time of the electron, given by, $t_B^e = 2.846 \times 10^{-22}$s. This theory provides another central relationship between the length scale L_B and the quantum information entropy S_{VN} it contains which is a dimensionless quantity. which is given by Bassam's fundamental spatial equation, $L_B = (S_{VN}/\pi) \cdot \lambda_0$. (upon restoring SI units), where λ_0 represents the reduced Compton wavelength. We can obtain the Bassam quantum length of the electron, which represents the smallest possible distance that an electron can travel from one place to another, and which is given by the calculated value $L_B^{(e)} \approx 8.52 \times 10^{-14}$m. Through this theory we obtain another central relationship between the mass spectral scale $m_{B,n}$ and the quantum information entropy S_{VN} it contains, which is a dimensionless quantity, as given by Bassam's fundamental mass spectral equation. $m_{B,n} = m(1 + (2\pi^2/S_{VN})\, n)$, New particles can be obtained, namely excited electrons (excited Bassam electrons), with a mass of $m_B^{(e)} \approx 15.1\,\mathrm{MeV/c^2}$ They are obtained through the (Dirac–Bassam) equation via the decay channel $e_B^- \rightarrow e^- + \gamma$. The theory also explains the behavior of the photon, where $S_{VN} = 0$ and its characteristic time must be $t_B^{(Ph)} = 0$, which provides a quantum explanation for the relativistic freezing of time at the speed of light c This is attributed to the fact that the photon does not possess quantum information entropy therefore its speed is maximal. This theory also seeks to provide an interpretation by exploring the implications and predictions it generates for various physical scenarios. The existence of quantum time necessarily leads us to the existence of a corre-sponding quantum space, thereby paving the way toward the unification of space and time. Through this endeavor, we aim to contribute to the ongoing dialogue concerning the origin of time and the origin of space, and to address the question of where the three spatial dimensions and the fourth, temporal dimension come from. Since this theory derives all components of the universe from information, the universe becomes a network of indivisible quantum pixels. This testable theory, which naturally establishes a connection between time and quantum information, extends to addressing other problems in physics, This opens up wide-ranging prospects for future applications in research and development, as well as in both theoretical and experimental investigations, particularly in experiments at facilities such as **Belle II** and **LHCb**, within the framework of high-energy particle physics. This theory

presents for the first time and in accordance with the researcher's findings, these fundamental equations and results. Bassam's fundamental time equation, t_B , Bassam's fundamental spatial equation, L_B and Bassam's fundamental mass spectral equation, $m_{B,n}$. The quantized Bassam coordinates are $(t_B \cdot n_t , L_B \cdot n_x , L_B \cdot n_y , L_B \cdot n_z)$, which represent new physical equations and quantities that have not been proposed or discovered in any previous physical model.

2. THEORETICAL FRAMEWORK AND DERIVATION

Everything in this universe can be interpreted as a holographic projection of data and information encoded on two-dimensional surfaces [9]. In this case, we can establish a connection between two fundamental theories in modern theoretical physics Einstein's general theory of relativity, which is a theory of gravity, and quantum theory. Therefore we can study black holes obtained by scientists through solutions to Einstein's equation in general relativity, from the perspective of quantum information physics [10]. Let us start with the equivalence principle, which states that any accelerated system experiences an effect identical to that of a gravitational field [11,12]. In examining the equivalence principle, we observe forces that mimic gravitational forces and are directed downward. Carrying out the corresponding calculations yields the following equation.

$$F - mg = am \tag{1}$$

where F [N] represents the force, m[kg] the mass of the body, $g[\mathrm{m/s^2}]$ the gravitational acceleration, and a $[\mathrm{m/s^2}]$ the acceleration of the body. Thus the product of mass and acceleration, ma, is not equal to the force F alone, but rather equals the force minus the gravitational term, $F - mg$. This mathematical expression concisely encapsulates the equivalence principle [13]. According to the fundamental physical relationship between acceleration and velocity, the final velocity of the body can be expressed by the following equation.

$$v = g\,t \tag{2}$$

Where v [m/s] represents the velocity, and t [s] represents time. If time continues to progress over long periods, then equation (2) predicts a velocity $v > c$ exceeding the speed of light in vacuum $c = 3 \times 10^8 \mathrm{m/s}$, Therefore, equation (2) can be replaced by another form that preserves the constant speed of light [14,15]. In modern theoretical physics, we study accelerating reference frames by using the light cone. where time progresses upward along the vertical axis, while space extends along the horizontal axes. The point at the center, denoted by D, represents the present point, while everything above it represents the future and everything below it represents the past [8], as illustrated in Fig 1.

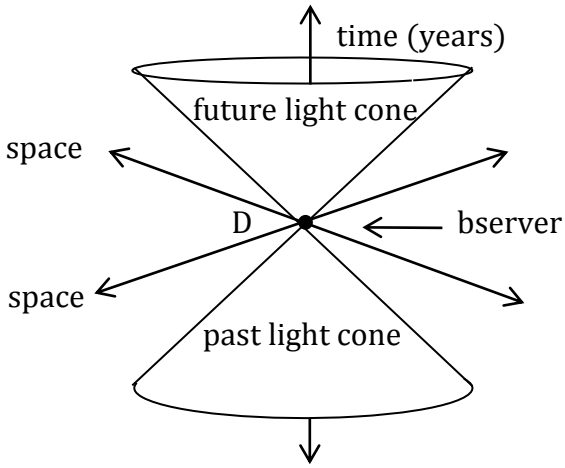


Fig. 1: Use of the Light Cone

Through the use of the light cone, accelerating reference frames can be represented, where light propagates at a 45-degree angle. Then, we express the slanted line denoted by H, which represents the speed of light $c = 1$. The acceleration of the body is represented by the curve denoted by X, which represents the accelerating reference frame and extends to infinity [16]. The point denoted by D represents the present.

In this case, no matter how much the body accelerates, it does not reach the speed of light. As the body accelerates through time, the line denoted by Y intersects with the line denoted

by X. The point denoted by Y_1 can be expressed as the point of intersection, as shown in Fig 2.

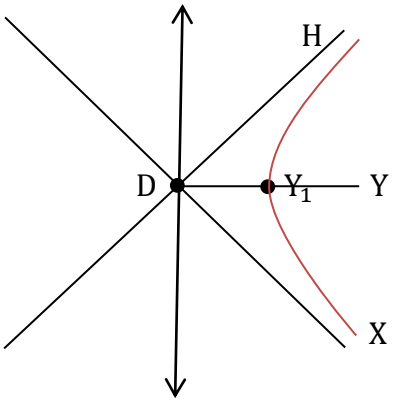


Fig. 2: Light cone and accelerating reference frames.

As the body accelerates through time, the line denoted by Y, which represents all the points of intersection with the curve denoted by X, changes from one intersection point to another. The intersection points can be denoted by the following symbols $Y_1Y_2Y_3$, as shown in Fig 3.

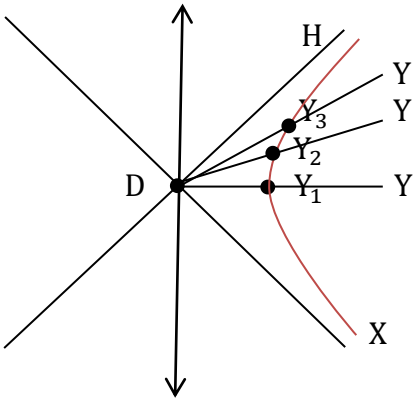


Fig. 3: Light cone and intersection points.

For the velocity of the body to become equal to the speed of light, the line denoted by Y becomes coincident with the slanted line denoted by H. However, it does not cross or intersect it, instead, the two lines become parallel in direction. At that point, the motion of the body becomes equal to the speed of light. This leads to time becoming infinite, therefore the slanted line denoted by H is referred to as the Horizon [17]. Every body that becomes behind the slanted line denoted by H, which represents the horizon, appears to accumulate at the horizon with respect to the accelerating reference frame, as shown in Fig 4.

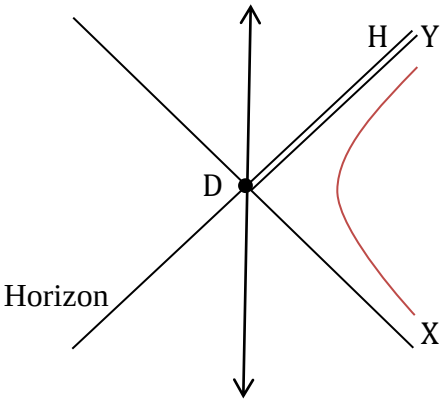


Fig 4: Coincidence of the line Y with the horizon H upon reaching the speed of light.

This horizon phenomenon is analogous to what occurs in black holes, where the horizon can be described as the surface composed of the data and information pertaining to all objects that fall into it [18]. The entropy of a system arises from all the transformations and changes occurring in the atomic particles constituting a given system. Consequently, the entropy of a black hole serves as a measure of the information about its interior, which is inaccessible to an external observer [19]. When light falls into a black hole, it cannot escape. Its mass increases with the addition of any radiation or matter. Consequently a small change in the system's entropy is accompanied by a corresponding change in its energy and mass, relative to the change in the area of the event horizon [20,21]. The simplest solution to Einstein's gravitational field equations in general relativity is the Schwarzschild radius [22]. If the entire mass of a black hole is compressed to the Schwarzschild radius, a black hole is formed from which light cannot escape. The Schwarzschild radius is given by the following equation.

$$r_s = \frac{2MG}{c^2} \tag{3}$$

where r_s [m] represents the Schwarzschild radius. Where M [kg] is the mass, $G = 6.674 \times 10^{-11}[\mathrm{m^3 kg^{-1} s^{-2}}]$ is the gravitational constant, and c [m/s] is the speed of light. The Bekenstein entropy represents the small amount

of information accumulating on the event horizon surface of a black hole to comply with the laws of black hole thermodynamics, as observed by an external observer [23]. Consider a black hole with the Schwarzschild radius [24], as shown in Fig 5.

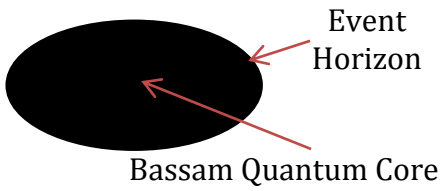


Fig. 5: A black hole with Schwarzschild radius r_s and the Bassam Quantum Core.

Anything falling onto the event horizon of a black hole can be expressed as a small amount of information. Taking an elementary particle, the smallest unit of information-for example, a photon-and letting it fall onto this horizon, it represents the information to be added to the black hole's horizon [23].Then, we assign to the photon $r_s = \lambda$ where λ [m] represents the wavelength of the photon, which is equal to r_s representing the Schwarzschild radius The energy of the photon falling onto the surface of the black hole is then considered through the Planck-Einstein relation, which is given by the following equation.

$$E = \frac{2\pi\hbar c}{\lambda} \tag{4}$$

where E [J] represents the photon energy, π is a mathematical constant. $\hbar = 1.0545718 \times 10^{-34}$ [J · s] is the reduced Planck constant. When a photon falls onto the event horizon of a black hole, the black hole's mass increases by a certain amount each time photon falls onto the event horizon of a black hole. This added mass represents a small amount of information [25]. Therefore, we obtain the following equation.

$$m = \frac{E}{c^2} = \frac{2\pi\hbar}{c\lambda} \tag{5}$$

where c^2 [m^2/s^2] represents the square of the speed of light. We then let another photon fall onto the event horizon of the black hole, and r_s which represents the radius of the black hole, increases by an amount each time a photon falls onto the event horizon [24]. Substituting equation (3) into equation (5) we obtain the following equation.

$$r_s = \frac{4\pi G\hbar}{c^3\lambda} \tag{6}$$

where c^3[m^3/s^3] represents the cube of the speed of light, and λ represents the wavelength of the photon. In particle physics, the photon is not continuous electromagnetic radiation but rather a discrete form, or what is termed a quantum. It consists of energy packets and can exhibit both wave-like and particle-like behavior [26]. Therefore, waves can be studied to probe atomic properties within the framework of atomic physics, thereby advancing our understanding of electromagnetic radiation. The photon consists of perpendicular electric and magnetic field components, which oscillate in space to constitute the photon [27]. The wavelength and the form of its oscillation define electromagnetic radiation, and the wave can oscillate in space to give rise to a field [28]. It is well established in the laws of quantum mechanics and relativity that linear momentum is a fundamental concept atomic particles and electromagnetic radiation all possess linear momentum, and the photon is no exception [29]. where the photon falls onto the event horizon of the black hole and behaves as a wave. The linear momentum equation of the photon, which expresses wave-particle duality [30], can be written as follows.

$$p = \frac{h}{\lambda} \tag{7}$$

where p represents the linear momentum of the photon, h is the Planck constant, and λ is the wavelength of the electromagnetic radiation. When a photon collides with a low-mass atomic particle, it can cause the particle to be deflected or even accelerated, thereby imparting additional momentum to it. The linear momentum law can be expressed by the following equation.

$$p = mv \tag{8}$$

where p [kg. m/s] is the linear momentum of the atomic particle, v is its velocity, and m is its

mass. The photon, as a quantum of information, falls onto the surface of the black hole and embodies the principle of wave particle duality, whereby atomic particles for example, electrons can exhibit both wave-like and particle-like behavior. Substituting Eq. (8) into Eq. (3), and setting the black hole mass M equal to the particle mass m, yields the following equation.

$$r_s = \frac{2pG}{vc^2} \tag{9}$$

In natural units, equation (9) simplifies to $r_s = 2p/v$, and equation (3) reduces to $r_s = 2m$. Equating these two expressions recovers equation (8), thereby confirming the consistency of the derivation. In particle physics, a simple law representing Einstein's mass-energy equivalence [31]. is given by the following equation.

$$E = mc^2 \tag{10}$$

Substituting equation (9) into equation (10) yields the following equation.

$$E = \frac{2pGm}{v.r_s} \tag{11}$$

For a slowly moving electron, its energy [5] can be expressed by the following equation.

$$E = \frac{p^2}{2m} \tag{12}$$

where p^2 [$\mathrm{kg^2.m^2/s^2}$] represents the square of the electron's linear momentum. Given that the electron exhibits wave-particle duality and that all atomic particles possess wave-like properties, by substituting equation (11) into equation (12), we obtain the following equation.

$$m^2 = \frac{r_s vp}{4G} \tag{13}$$

where m^2 [$\mathrm{kg^2}$] represents the square of the equivalent mass. A given field can be studied mathematically to investigate waves. The simplest form of such a field in which waves are studied is a one-dimensional field, where the wave propagates periodically [32], as shown in Fig 6.

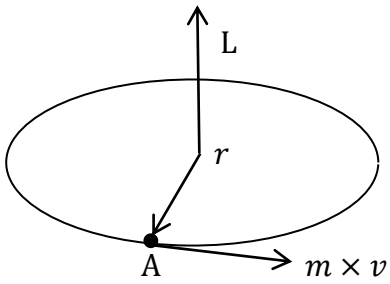


Fig. 6: Representation of a One-Dimensional Field for Studying Waves.

Through mathematical representation, we can eliminate infinity. As we know in this one-dimensional circular field where these waves, representing photons radiation or atomic particles, oscillate the wave propagates indefinitely in the same direction without changing its direction. At this point, we can preserve the principle of conservation of linear momentum [33]. Through the one-dimensional field , we can study waves that represent electromagnetic radiation and carry specific, discrete amounts of energy. The wave is also given by the following equation.

$$\lambda = \frac{L}{n} \tag{14}$$

where L [m] represents the circumference of the circle, and $n = 0, \pm1, \pm2, \pm3, \ldots$ is a dimensionless integer that denotes the number of possible waves propagating along the circumference of the one-dimensional circle. Now, the linear momentum [34] is given by the following equation.

$$p = \frac{2\pi n\hbar}{L} \tag{15}$$

Substituting equation (13) into equation (15) yields the following equation.

$$m^2 = \frac{\hbar n v r_s \pi}{2GL} \tag{16}$$

where π is a mathematical constant, and L represents the circumference of the circle, given by $L = 2\pi r$. Assuming that $r_s = r$ and that $v = c$, equation (16) can be rewritten to obtain the following equation.

$$m^2 = \frac{\hbar nc}{4G} \tag{17}$$

where m^2 [kg^2] represents the square of the quantized mass. Stephen Hawking, in his study of black holes, was able to derive the entropy relation [35], given by the following equation.

$$S = \frac{4\pi k_B G m^2}{\hbar c} \tag{18}$$

where S [J/K] represents the quantum gravitational entropy, and k_B [J/K] is the Boltzmann constant. Substituting equation (17) into equation (18) yields the following equation.

$$S = \pi\, k_B n \tag{19}$$

Mathematical equations can be used in a special system of units, where we choose suitable units to make the speed of light dimensionless , and we can set $c = 1$ (in natural units). In this case, we obtain an important result in particle physics mass is equivalent to energy, and the two are equivalent concepts in particle physics [36]. For example, when an electron meets a positron, its antiparticle, they annihilate each other, producing high-energy electromagnetic radiation. The resulting energy is calculated to be precisely the sum of the masses of the electron and the positron. Hence, in particle physics, mass and energy are treated as a single equivalent concept where $E = m$ represents the energy of atomic particles that possess mass Another constant can be expressed in natural units, where the gravitational constant is set to $G = 1$, given that gravity is weak in particle physics and can often be neglected Similarly, another constant can be set in natural units, namely the Boltzmann constant $k_B = 1$, to simplify the mathematical formulation [37]. Consequently, equation (19) can be rewritten to obtain the following equation.

$$S = \pi\, n \tag{20}$$

where S represents the quantized black hole entropy, which is a dimensionless quantity, In quantum physics, the electron is described by a wave function, as indicated by Schrödinger in his famous equation, which establishes a relationship describing the electron's rotation and motion around the nucleus of the atom [4]. The electron possesses a specific wavelength and frequency. Thus, its wave properties can be studied and represented using various periodic functions, or in terms of the fundamental trigonometric functions, namely sin and cos . Exponential functions can also be expressed in complex form, such as e^{ikx}, and all such mathematical functions can be utilized for the mathematical formulation of wave physics [38]. In the study of a complex number, which represents a combination of real and imaginary parts, it consists of two components. the real part along the axis denoted by X, and the imaginary part along the axis denoted by Y, representing the imaginary number. It is an arrow, denoted by u, which has a specific magnitude and a specific direction, and possesses an angle denoted by φ. The arrow u originates from the point denoted by Q , which represents the intersection of the (X,Y) axes. When the angle is $(\varphi = 0)$, the arrow u coincides exactly with the X axis, which represents the real number. If, on the other hand, the angle is given by $(\varphi \neq 0)$, then the number possesses two components a real part and an imaginary part. Moreover, the angle φ can assume the values $\pi, 2\pi, 4\pi$, and it can exhibit periodic behavior, as illustrated in Fig 7.

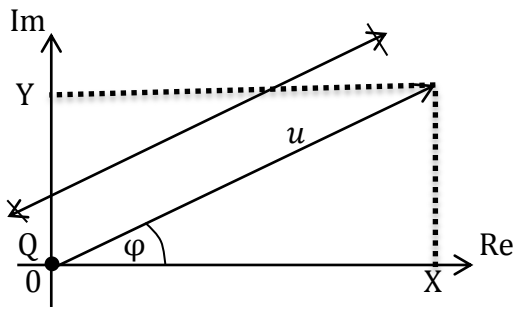


Fig. 7: Representation of Imaginary (Complex) Numbers

The complex number, represented as an arrow and denoted by w, It represents the change along the circumference of the circle. It has two projections. Projection along the real axis denoted by X We can then observe from the trigonometric relationship that the real projection of the complex number can be found in the following form $cos\,\varphi$ and the imaginary part, which lies along the axis denoted by u It

can be written in the form $sin\ \varphi$, as illustrated in Fig 8.

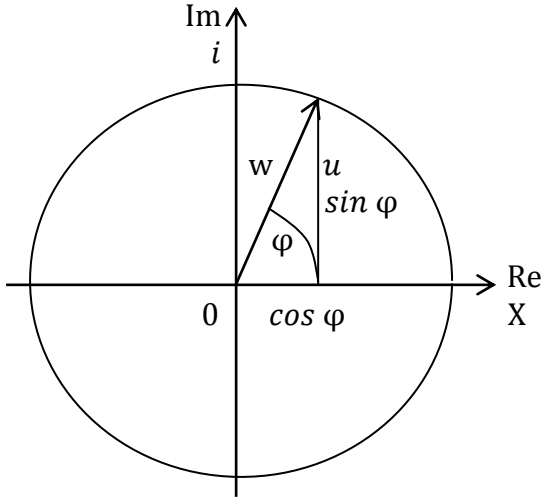


Fig. 8: Representation of Complex Numbers

Therefore, a complex number can be written in the form $cos\ \varphi + isin\ \varphi$, which represents a type of imaginary periodic function. where the symbol i represents the imaginary unit. We can now write the relation representing the periodic functions in the mathematical representation of waves [39], as follows.

$$e^{i\varphi} = cos\ \varphi + isin\ \varphi \tag{21}$$

Periodic functions can be represented mathematically in the form e^{ikx} , It represents the wave that is periodic along the circumference of the circle, denoted by L which lies in a one-dimensional domain, where i represents the imaginary unit. where k [m^{-1}] represents the wavenumber. x It represents the position along the one-dimensional domain over which the wave propagates along its circumference. For the wave to constitute a periodic function, it can start from point A, propagate along the circumference of a circle, and return to A. It then undergoes subsequent cycles, advancing and returning to its starting point without any change. where the wave propagates along the circle through one complete cycle, as illustrated in Fig 6. Therefore, we can obtain relations that describe the wave and its propagation in order for it to become a periodic function. The following relations can be obtained.

$$e^{ikL} = e^{i2\pi n} \tag{22}$$

Simple mathematical relations describing wave propagation [40] in a one dimensional system can be obtained, yielding the following equations.

$$k = \frac{2\pi n}{L} = \frac{2\pi}{\lambda} = \frac{P}{\hbar} \tag{23}$$

The linear momentum can be expressed in the following form $k = p$ where we use (in natural units) in which $\hbar = 1$ The wave propagating along the circumference of the circle has a wave number equal to its linear momentum. The photon energy, associated with electromagnetic radiation, can likewise be expressed by the following equation.

$$E = \hbar\omega \tag{24}$$

where ω represents the angular frequency of the wave [41]. The energy can be expressed as the angular frequency when $\hbar = 1$, corresponding to the relation $E = \omega$. Using the simple Dirac form [5,42]. yields the following equation.

$$\frac{\partial\psi}{\partial t} - \frac{\partial\psi}{\partial x} = 0 \tag{25}$$

The term $\partial\psi/\partial t$ represents the derivative of the quantum wave function with respect to time and is equal to $-i\omega\psi$, where ω is the angular frequency, corresponding to the energy via $E = \omega$. The other term, $\partial\psi/\partial x$, represents the derivative with respect to space and is equal to $ik\psi$, where k is the wavenumber, corresponding to the linear momentum via $p = k$. The wave function ψ is a quantum wave function and can be a positive-valued function with a quantized angular frequency, given by the form $e^{i(kx-\omega_n t)}$.where ω_n is the quantized angular frequency, which depends on the integer n. and it is the number of possible waves propagating along the circumference of the circle within a one-dimensional field. As previously mentioned, the field under consideration is one-dimensional. Consequently, a periodic wave function is obtained, which is given by the following equation.

$$\omega_n = k \tag{26}$$

We can represent atomic particles moving to the right and atomic particles moving to the left If the sign is positive in this case, the atomic particles are moving to the right. We then add the symbol R to indicate the rightward direction [43]. herefore, equation (25) can be rewritten to yield the following equation.

$$\frac{\partial \psi_R}{\partial t} + \frac{\partial \psi_R}{\partial x} = 0 \tag{27}$$

However, if the wave has a negative sign, then the atomic particles are moving to the left. Then, the symbol L is added to denote the left-hand side [43]. Therefore, equation (25) can be written to yield the following equation.

$$\frac{\partial \psi_L}{\partial t} - \frac{\partial \psi_L}{\partial x} = 0 \tag{28}$$

As we know, we now have two different equations: one represents the atomic particles moving to the right, as given by Equation (27), while the other represents the atomic particles moving to the left, as given by Equation (28) To obtain a mathematical form that represents both Equations (27) and (28) in a single equation, we must introduce a matrix denoted by α, which is also of the simple type and is given by the following equation .

$$\alpha = \begin{matrix} 1 & 0 \\ 0 & -1 \end{matrix} \tag{29}$$

where the matrix denoted by α is either positive or negative. If positive, the atomic particles are directed to the right. if negative, they are directed to the left [44]. Therefore, equation (25) can be written to yield the following equation.

$$\frac{\partial \psi}{\partial t} - \alpha \frac{\partial \psi}{\partial x} = 0 \tag{30}$$

Now, equation (30), which represents the sum of equations (27) and (28), can be written to yield the following equation.

$$\omega_n = -\alpha\, k \tag{31}$$

By substituting Equation (31) into Equation (20), we obtain the following equation.

$$\omega_{(\frac{S}{\pi})} = -\alpha\, k \tag{32}$$

where S represents the quantized black hole entropy. In Equation (20), where n is an integer representing the number of possible waves propagating along the circumference of the one-dimensional circle This interpretation provides a wave-like and geometric origin. In the previous model, we assumed the incidence of a single photon, which represents a small quantum of information to be added to the event horizon of the black hole. where the photon behaves as a wave, and all atomic particles exhibit wave-like behavior, demonstrating wave-particle duality , Thus, they can behave either as particles or as waves. Since S does not depend on specific properties of the black hole and is a dimension-less quantity, the relation can be generalized to include any physical system (including atomic particles). Therefore, S can be replaced by S_{VN}, where S_{VN} represents the quantum information entropy (von Neumann entropy). and it is also a dimensionless quantity (in natural units, $\hbar = k_B = c = 1$), so that the new framework encompasses it. Just as Einstein expanded the concept of gravity by replacing the Newtonian density with the energy-momentum tensor $T_{\mu\nu}$ within the framework of his general theory of relativity, we seek to extend relation (32) so that it applies to any quantum system possessing quantum information entropy (von Neumann entropy), which is a dimensionless quantity. Therefore, the following equation can be obtained.

$$\omega_{(\frac{S_{VN}}{\pi})} = -\alpha\, k \tag{33}$$

π is a mathematical constant. The energy-momentum equation [45] can now be written, given by the following equation .

$$\omega = k + m \tag{34}$$

where Equation (34) contains the term represent-ing the mass denoted by m. However, Equation (33) does not contain a term representing mass. Therefore, a term representing the mass, namely $m\beta$, can be added [5]. where m rep-resents the mass, and β represents a matrix,

which is also of the simple type, as given by the following equation.

$$\beta = \begin{matrix} 1 & 0 \\ 0 & 1 \end{matrix} \quad (35)$$

The term $m\beta$ can now be added to Equation (33) to yield the following equation.

$$\omega_{(\frac{S_{VN}}{\pi})} = -\alpha\, k + m\beta \quad (36)$$

Equation (34) can be squared to yield the following equation.

$$\omega^2 = k^2 + m^2 \quad (37)$$

Equation (36) can be squared to yield the following equation.

$$\omega^2{}_{(\frac{S_{VN}}{\pi})} = \alpha^2\, k^2 + m^2\beta^2 + mk(\alpha\beta + \beta\alpha) \quad (38)$$

where the term $\omega^2{}_{(S_{VN}/\pi)}$ in Equation (38), with S_{VN} representing the quantum information entropy (von Neumann entropy), which is a dimensionless quantity. π is a mathematical constant. It corresponds to the term ω^2 in Equation (37). And the term $\alpha^2\, k^2$ in Equation (38) corresponds to the term k^2 in Equation (37). where α^2 represents a square matrix $(\alpha^2 = 1)$, given by the following equation.

$$\alpha^2 = \begin{matrix} 1 & 0 \\ 0 & 1 \end{matrix} \quad (39)$$

The term $m^2\beta^2$ in Equation (38) corresponds to the term m^2 in Equation (37), where β^2 represents a square matrix $(\beta^2 = 1)$ [5], given by the following equation.

$$\beta^2 = \begin{matrix} 1 & 0 \\ 0 & 1 \end{matrix} \quad (40)$$

As for the term $mk(\alpha\beta + \beta\alpha)$ in Equation (38), it is absent and does not correspond to Equation (37). Therefore, it must be eliminated by means of $(\alpha\beta + \beta\alpha)$. Through the laws of matrix multiplication the term $\alpha\beta$ can be deduced, as given by the following equation.

$$\alpha\beta = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \quad (41)$$

Similarly, the term $\beta\alpha$ can be derived through the laws of matrix multiplication, as given by the following equation.

$$\beta\alpha = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad (42)$$

The values of the two given expressions in Equations (41) and (42), which appear in Equation (38), can be determined. which are contained within the term $mk(\alpha\beta + \beta\alpha)$ and which appear in the form $(\alpha\beta + \beta\alpha)$. Upon adding the two matrices together, a zero matrix is obtained, which takes the form $\alpha\beta + \beta\alpha = 0$ [5,46], as given by the following equation.

$$\alpha\beta + \beta\alpha = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} + \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = 0 \quad (43)$$

In this case, the researcher was able to introduce S_{VN}, which represents the quantum information entropy (von Neumann entropy) and is a dimensionless quantity. Moreover, π is a mathematical constant. Equation (38) can now be written after eliminating the term $mk(\alpha\beta + \beta\alpha)$ due to the identity $\alpha\beta + \beta\alpha = 0$, to yield the following equation.

$$\omega_{(\frac{S_{VN}}{\pi})}{}^2 = \alpha^2\, k^2 + m^2\beta^2 \quad (44)$$

We can write the relative natural time relation, which is given by the following equation.

$$\tau = \frac{t}{t_0} \quad (45)$$

where τ represents the characteristic natural time of the system (a dimensionless quantity), t represents the physical time, and t_0 represents the natural time scale of the system.

2. 1 BASSAM'S FUNDAMENTAL QUANTUM EQUATION OF TIME

The term $\partial\psi/\,\partial\mathrm{t}$ in Equation (25) represents the derivative of the quantum wave function with respect to time and is equal to $-i\omega\psi$, which

gives $\psi(t)$ [47]. Therefore, the term $\omega_{(S_{VN}/\pi)}$ in Equation (36) becomes in the following form $\partial\psi/\partial t\,\big|_{t_B=S_{VN}/\pi}$ The relation $t_B = \tau$ can be obtained (in natural units). The following equation can now be written.

$$\tau = \frac{S_{VN}}{\pi} \tag{46}$$

After the researcher obtained Equation (46), substituting it into Equation (45) yields Bassam's fundamental time equation. which is denoted by the symbol t_B (where we add the letter B, which indicates the researcher's designation). and it represents Bassam's quantum time, associated with S_{VN}, which represents the quantum information entropy (von Neumann entropy) and is a dimensionless quantity, In addition, π is a mathematical constant, and t_0 represents the natural time scale of the system. Upon restoring SI units [48], Bassam Muwaffaq Salman Abdul Karim was able to derive Bassam's fundamental time equation, which is given by the following equation.

$$t_{B_{assam}} = \frac{S_{VN}}{\pi} \cdot t_0 \tag{47}$$

where S_{VN} represents the quantum information entropy (von Neumann entropy) for any quantum system, as given by the following equation.

$$S_{VN} = -k_B\, Tr(\rho \ln \rho) \tag{48}$$

In Equation (48), where S_{VN} [J/K] represents the quantum information entropy (von Neumann entropy). k_B is the Boltzmann constant. In our derivations, π, a mathematical constant, appears and is eliminated from Equation (20). Upon transitioning to natural units, $k_B = 1$. Consequently, the entropy becomes dimensionless. When SI units are employed, preserving dimensional consistency requires the inclusion of k_B in Equation (47). Thus, k_B is algebraically canceled from the numerator and the denominator, which ensures the consistency of Equation (47). where S_{VN} represents the quantum information entropy (von Neumann entropy), and it is a dimensionless quantity. ρ represents the density matrix, which describes the quantum state of the system. Tr represents the trace, which is the sum of the diagonal elements of the matrix. $\rho \ln$ denotes the logarithm of the density matrix. Equation (47) can be written (upon restoring SI units). With k_B present in the numerator and denominator, Bassam's fundamental time equation can be written as given by the following equation.

$$t_B = \frac{S_{VN}^{(\mathrm{SI})}}{k_B \pi} \cdot t_0 \tag{49}$$

where $S_{VN}^{(SI)}$ represents the quantum information entropy (von Neumann entropy) with dimensions [J/K] (upon restoring SI units). Equation (47) can be written, when using natural units with $t_0 = 1$, to yield the following equation.

$$t_B = \frac{S_{VN}}{\pi} \tag{50}$$

We can write the fundamental relation connecting length, time, and velocity, as given by the following equation.

$$L = v \cdot t \tag{51}$$

where L [m] represents the distance, v represents the velocity, and t represents the time. When the velocity is the same as the speed of light c, we can consider $t_B = t$, and we obtain the following equation

$$L_B = c \cdot t_B \tag{52}$$

Substituting Equation (52) into Equation (47) yields Bassam's fundamental spatial equation. which is denoted by the symbol L_B, representing the Bassam quantum length. The researcher was able to obtain Bassam's fundamental spatial equation, which is given by the following equation.

$$L_{B_{assam}} = \frac{S_{VN}}{\pi} \cdot \lambda_0 \tag{53}$$

where S_{VN} represents the quantum information entropy (von Neumann entropy), and it is a dimensionless quantity. where λ_0 [m] represents the reduced Compton wavelength. and it is given by the formula $\lambda_0 = \hbar/mc$. Equation (53)

can be written, when using the International System of Units, as the following equation.

$$L_B = \frac{S_{VN}^{(\mathrm{SI})}}{k_B \pi} \cdot \lambda_0 \qquad (54)$$

where $S_{VN}^{(\mathrm{SI})}$ represents the quantum information entropy (von Neumann entropy) with dimensions [J/K] (when using the International System of Units). A complete correspondence can be written between Equation (47), which is Bassam's quantum time equation, and Equation (53), which is Bassam's quantum length equation (in natural units, $\hbar = k_B = c = 1$), Since time and space are the same entity, two sides of the same coin, the following equation can be written.

$$t_B = L_B = \frac{S_{VN}}{\pi} \cdot \frac{1}{m} \qquad (55)$$

Contrary to what has been established in classical and relativistic physics. where the space and time coordinates, denoted by the symbols (*x, y, z, t*), are treated as underived. We use them without the model answering about the origin of length itself or time itself. In Bassam's Unified Quantum Theory, the first derivation of the origin of these four components is presented. And from where do these three spatial coordinates and the fourth coordinate, time, come. Within the framework of this theory, length is not given a priori. but rather emerges from fundamental pixels, which is the Bassam quantum length L_B. as given by the formulas $L_B \cdot n_x = x$, $L_B \cdot n_y = y$, $L_B \cdot n_z = z$ time is not given a priori. but rather emerges from fundamental pulses, which is the Bassam quantum time t_B. as given by the formula $t_B \cdot n_t = t$. Accordingly, the four coordinates are redefined. The coordinate x is not merely a number, but rather it is $L_B n_x$. The coordinate y is $L_B n_y$, the coordinate z is $L_B n_z$, and the coordinate t is $t_B\, n_t$. where (n_t, n_x, n_y, n_z) represent integers that count the spacetime steps. Their values, t_B and L_B, are derived from S_{VN} , the quantum information entropy (von Neumann entropy), which is a dimensionless quantity; π is a mathematical constant, and m is the mass. and that is through the fundamental equations of the theory. which turns the coordinates from tools of description into physical quantities that can be termed Bassam's quantized coordinates, given by their general form $(t_B \cdot n_t\, , L_B \cdot n_x\, , L_B \cdot n_y\, , L_B \cdot n_z)$. And since the researcher views these coordinates as a discrete (quantized) network that assists in formulating all the laws of modern theoretical physics, They can be written as Bassam coordinates in this simple form, $(t_B\, , x_B\, , y_B, z_B)$.

2. 2 DIRAC–BASSAM EQUATION

After the researcher was able to introduce S_{VN}, which represents the quantum information entropy (von Neumann entropy), and adding π, which is a mathematical constant, to Equation (44), and as we know, the field we studied is one-dimensional. Now, the Dirac–Bassam equation can be written on the axes (x, y, z, t), as in the following equation.

$$i \frac{\partial \psi}{\partial t}\Big|_{t=t_{B_{assam}}} = -i\alpha_n \frac{\partial \psi}{\partial x_n} + m\,\beta\psi \qquad (56)$$

where i represents the imaginary unit, and the term $\partial\psi/\partial t$ represents the derivative of the quantum wave function with respect to time, The vertical line represents the constraint condition, and t_B represents the Bassam quantum time. π is a mathematical constant, and the symbols α_n, where α is the Dirac matrix and the symbol n represents $(\alpha_1, \alpha_2, \alpha_3)$, and the term $\partial\psi/\partial x_n$ represents the derivative of the quantum wave function with respect to space, and the symbols x_n, where x represents space and $n = 1,2,3$ representing the axes (x, y, z), and m represents mass, and β is a Dirac matrix, and ψ is the quantum wave function.

3. RESULTS AND APPLICATIONS

3. 1 FUNDAMENTAL EQUATIONS

This theory presents Equation (47), which is Bassam's fundamental time equation, associated with any physical system through its own quantum information entropy (in natural units, $\hbar = k_B = c = 1$), and which is given by the following equation.

$$t_B = \frac{S_{VN}}{\pi} \cdot \frac{1}{m} \quad (57)$$

where t_B represents Bassam's quantum time. , $1/m = t_0$ (the natural time scale of the system), S_{VN} represents the quantum information entropy (von Neumann entropy) for any quantum system, and it is a dimensionless quantity, which is given by the following equation.

$$S_{VN} = -Tr(\rho \ln \rho) \quad (58)$$

3. 2 APPLICATION OF BASSAM'S FUNDAMENTAL TIME EQUATION TO THE ELECTRON

As its first and novel application case, we study an electron system in a maximally mixed quantum state (unpolarized spin). The quantum information entropy (von Neumann entropy) is $S_{VN}^{(e)} = \ln 2 = 0.693$, and the electron mass is $m_e = 0.511\ \mathrm{MeV/c^2}$.

$$t_0^{(e)} = \frac{1}{m_e} = 1.29 \times 10^{-21}\mathrm{s} \quad (59)$$

Substituting $1/m_e$ into Equation (59) yields the internal Bassam quantum time of the electron system.

$$t_B^{(e)} = \frac{S_{VN}^{(e)}}{\pi} = \frac{\mathrm{In}2}{\pi} = \frac{\mathbf{0.693}}{3.1416} = 0.221\ [\frac{1}{m_e}] \quad (60)$$

By converting the value (upon restoring SI units).

$$t_B^{(e)} = 0.221 \times 1.288 \times 10^{-21} \approx 2.846 \times 10^{-22}\mathrm{s}$$

yielding a value of $t_B^e \approx 2.846 \times 10^{-22}$s, which represents the internal Bassam quantum time of the electron in the maximally mixed state.

3. 3 APPLICATION OF BASSAM'S FUNDAMENTAL SPATIAL EQUATION TO THE ELECTRON

Upon applying Equation (53), which is Bassam's fundamental spatial equation, associated with any physical system through its own quantum information entropy (upon restoring SI units), where $L_B^{(e)}$ represents the Bassam quantum length of the electron, which is given by the following equation.

$$L_B^{(e)} = \frac{S_{VN}^{(e)}}{\pi} \cdot \frac{\hbar}{m_e c} \quad (61)$$

$L_B^{(e)}$ represents the Bassam quantum length of the electron. $S_{VN}^{(e)}$ represents the quantum information entropy (von Neumann entropy) of the electron, and it is a dimensionless quantity. Upon numerical substitution.

$$L_B^{(e)} = 0.220636061 \times 3.861592676 \times 10^{-13} \approx 8.51998 \times 10^{-14}\mathrm{m}$$

We obtain a value of $L_B^{(e)} \approx 8.52 \times 10^{-14}$m, which represents the Bassam quantum length of the electron in the maximally mixed state-the smallest quantum distance that the electron can traverse from one place to another.

3. 4 APPLICATION OF BASSAM'S FUNDAMENTAL TIME AND SPATIAL EQUATIONS TO THE PHOTON

As a second and new application case (upon restoring SI units), we study the photon in a pure (polarized) state, where the quantum information entropy (von Neumann entropy) is $S_{VN}^{(Ph)} = 0$. Substituting into Equation (47) yields the Bassam quantum time of the photon, ($t_B^{(Ph)} = 0$). Upon applying Equation (53), we obtain the Bassam quantum length of the photon, ($L_B^{(Ph)} = 0$). This result, which causes the photon to travel at the speed of light, is that the vanishing of its entropy, $S_{VN}^{(Ph)} = 0$, leads to the vanishing of its time ($t_B^{(Ph)} = 0$) and its space ($L_B^{(Ph)} = 0$), thereby allowing the attainment of the maximum speed.

3. 5 SUMMARY OF NUMERICAL RESULTS

System	S (von Neumann entropy)	natural time scale t_0(s)	Bassam time t_B(s)	Bassam length L_B(m)
Electron, pure spin state	ln 2	0.221	2.846×10^{-22}	8.520×10^{-14}
Photon, pure polarization state	0	-	0	0

3. 6 VERIFICATION OF THE FUNDAMENTAL EQUATIONS OF THE THEORY AND RESTORATION OF THE CONSTANT SPEED OF LIGHT

Within the framework of this theory, the speed of light c acquires a new physical meaning, it is no longer a primary postulate as in special relativity. Rather, it has become a necessary consequence derived from the fundamental laws of Unified Quantum Theory. By applying these laws, the Bassam quantum time of the electron is given as $t_B^{(e)} \approx 2.846 \times 10^{-22}$s. and the Bassam quantum length of the electron, $L_B^{(e)} \approx 8.51998 \times 10^{-14}$m. And by dividing these two fundamental quantities, we obtain the value of the speed of light constant, which confirms the exact numerical agreement with the standard value of the speed of light. yielding $v = c$, since the speed of light emerges naturally from the lattice spacetime structure. Through the derivational verification by dividing Equation (47) and Equation (53), we obtain the speed of light c, which mathematically proves the internal consistency of the theory with special relativity. Therefore, the speed of light can be obtained through the following equation.

$$c = \frac{L_B}{t_B} \tag{62}$$

3. 7 PREDICTIONS FOR THE EXISTENCE OF EXCITED ELECTRONS (BASSAM ELECTRON)

By solving the Dirac-Bassam equation, which is given by Equation (56), And in the rest frame, where $P = 0$, this leads to time quantization, where $t_B \cdot n_t = t$, and through quantum mechanics [49], If time is quantized, then the energy takes discrete values given by the following equation.

$$E_n = \frac{2\pi\hbar}{t_B} n \tag{63}$$

where n $[0,1,2,3, \dots]$ is an integer taking non-negative values to describe excited particles. the excited Bassam electron, (e_B^-). the excited Bassam positron, (e_B^+) . It yields the positive mass via the negative-energy solutions of Equation (56), consistent with matter-antimatter symmetry. (upon restoring SI units), where t_0, which represents the natural time scale of the system, is given by the following equation.

$$t_0 = \frac{\hbar}{mc^2} \tag{64}$$

We can write Equation (47), which is Bassam's fundamental time equation, and which is given by the following equation.

$$t_B = \frac{\ln 2}{\pi} \cdot \frac{\hbar}{mc^2} \tag{65}$$

Upon substituting Equation (63) into Equation (65), we obtain the excitation energy, which is given by the following equation.

$$E_n = \frac{2\pi\hbar}{\frac{\ln 2}{\pi} \cdot \frac{\hbar}{mc^2}} n = \frac{2\pi^2}{ln\,2} mc^2 n \tag{66}$$

In addition to the rest energy mc^2, the total energy becomes as given by the following equation.

$$E_n^{(total)} = mc^2 \left(1 + \frac{2\pi^2}{\ln 2} n\right) \tag{67}$$

By dividing Equation (67) by c^2, we obtain Bassam's fundamental mass spectral equation in

its general form, which is given by the following equation.

$$m_{B_{assam},n} = m\left(1 + \frac{2\pi^2}{S_{VN}}n\right) \qquad (68)$$

Now, Equation (68), which is given by the following equation, can be applied to an electron.

$$m_{B,n} = m_e\left(1 + \frac{2\pi^2}{\ln 2}n\right) \qquad (69)$$

where $m_{B,n}$ represents the Bassam quantum mass spectrum. m_e represents the electron mass. S_{VN}, which represents the quantum information entropy (von Neumann entropy) and is a dimensionless quantity. Upon applying Equation (69) in the case $n = 0$, where $m_e = 0.511\ \mathrm{MeV/c^2}$, which is the mass of the ordinary electron. As for the case where $n = 1$, the equation yields a new particle that can be provisionally termed the excited Bassam electron. which is denoted by the symbol (e_B^-) , and which is given a mass of $m_B^{(e)} \approx 15.062\ \mathrm{MeV/c^2}$, as in the following equation.

$$m_B^{(e)} = m_e\left(1 + \frac{2\pi^2}{\ln 2}\right) \approx 15.063\ \mathrm{MeV/c^2} \quad (70)$$

This particle represents the first excited state in the Bassam quantum mass spectrum. The (e_B^-) can be detected experimentally in particle colliders. particularly in the Belle II experiment [50]. at **KEK** in Japan, via the decay channel $e_B^- \to e^- + \gamma$. The experimental signature appears as a resonance peak in the invariant mass distribution of the $(e\gamma)$ pair at a value of $m_B^{(e)} \approx 15.06\ \mathrm{MeV/c^2}$ (or approximately $m_B^{(e)} \approx 15.1\ \mathrm{MeV/c^2}$, with a natural width of $\Gamma \sim 0.1\ \mathrm{MeV}$ arising from the short lifetime of the particle. Moreover, this particle can be searched for in the **LHCb** experiment at **CERN** via pp collisions. or from reanalyzing existing data from the **BaBar** experiment at **SLAC** and the **LEP** experiment at **CERN** in the low-mass region, which has not been examined with sufficient precision in previous analyses [51].

3. 8 TABLE OF RESULTS FOR THE EXCITED BASSAM ELECTRON

Value	Symbol	Quantity
$\approx 15.063\ \mathrm{MeV/c^2}$	$m_B^{(e)}$	Bassam electron mass
$\approx 15.1\ \mathrm{MeV/c^2}$	$m_B^{(e)}$	Approximate value
$e_B^- \to e^- + \gamma$	-	Decay channel
$\sim 0.1\ \mathrm{MeV}$	Γ	Natural width
LHCb (CERN), Belle II (KEK, Japan), BaBar (SLAC), LEP (CERN), NA64 (CERN), PADME (INFN-LNF, Italy), DarkLight (ARIEL/Jefferson Lab), MEG II (PSI), Mu3e (PSI, Switzerland), SHiP (CERN), BDX (Jefferson Lab, United States), A1@Mainz (MAMI, Germany)	-	Proposed experiments and data analyses

3. 9 QUANTIZED EINSTEIN-BASSAM EQUATION FOR ENERGY AND MASS

For the integrated formulation in this theory, we can quantize the wavenumber. We assume the quantization of space in the form $L_B \cdot n_x = x$, as in the following equation.

$$k_n = \frac{2\pi}{L_B}\, n \qquad (71)$$

In the case $n = 1$, we obtain Bassam's equation for the wavenumber, which is given by the following equation.

$$k_{B_{assam}} = \frac{2\pi}{L_B} \quad (72)$$

where $k_{B_{assam}}$ is the Bassam wavenumber. upon substituting equation (71) into Equation (53), we obtain the following equation.

$$k_{B_{assam}} = \frac{2\pi^2}{S_{VN}} \cdot \frac{mc}{\hbar} \quad (73)$$

Upon applying Equation (72) to an electron, noting that the Bassam quantum length of the electron is $L_B^{(e)} \approx 8.52 \times 10^{-14}\text{m}$, therefore Therefore, the Bassam wavenumber for the electron is obtained from the following equation

$$k_{B_{assam}}^{(e)} = \frac{2\pi}{L_B^{(e)}} \approx 7.37 \times 10^{13}\ \text{rad/m} \quad (74)$$

The general law of quantized frequency can be written. Expressing the quantized time in the form $t_B \cdot n_t = t$, we obtain the following equation**.**

$$\omega_n = n \cdot \frac{2\pi}{t_B} \quad (75)$$

In the case $n = 1$, we obtain Bassam's equation for the angular frequency, which is given by the following equation

$$\omega_{B_{assam}} = \frac{2\pi}{t_B} \quad (76)$$

where $\omega_{B_{assam}}$ represents the Bassam angular frequency. By substituting Equation (76) into Equation (47), the following equation is obtaine.

$$\omega_{B_{assam}} = \frac{2\pi^2}{S_{VN}} \cdot \frac{mc^2}{\hbar} \quad (77)$$

Upon applying Equation (77) to an electron in its maximally mixed state, $S_{VN}^{(e)} = \ln 2$. We obtain the Bassam angular frequency for the electron, which is given by the value $\omega_{B_{assam}}^{(e)} \approx 2.21 \times 10^{22}\text{rad/s}$. In this context, we can write the quantized excitation dispersion relation in natural units, which is called the Bassam quantized dispersion equation, and which is a quantum extension of the dispersion relation in special relativity for excitation, given by the following equation.

$$\omega_{B_{assam,n}}^2 = k_{B_{assam,n}}^2 \quad (78)$$

where $\omega_{B_{assam,n}}^2$ and $k_{B_{assam,n}}^2$ are quantized quantities depending on the integer n. We can now write the energy, momentum, and mass equation in special relativity $E^2 = p^2c^2 + m^2c^4$. When $p = 0$, we can now write the quantized Einstein-Bassam equation for the total energy at rest, given by the following equation.

$$E_{B,n} = m_{B,n}\, c^2 \quad (79)$$

where $E_{B,n}$ is the total quantized energy, and $m_{B,n}$ is the total quantized mass. Now we can write the expanded form of the quantized Einstein-Bassam equation for energy and mass at rest in the International System of Units, which is given by the following equation.

$$E_{B,n} = mc^2\left(1 + \frac{2\pi^2}{S_{VN}} n\right) \quad (80)$$

3. 10 TABLE OF RESULTS: BASSAM ANGULAR FREQUENCY AND BASSAM WAVENUMBER

Unit	Value	Symbol	Quantity
rad /s	$\approx 2.210 \times 10^{22}$	$\omega_{B_{assam}}^{(e)}$	Bassam angular frequency (electron)
rad /m	$\approx 7.37 \times 10^{13}$	$k_{B_{assam}}^{(e)}$	Bassam wave number (electron)

4. DISCUSSION

4. 1 INTERPRETATION OF RESULTS

This theory presents Equation (47), termed Bassam's fundamental time equation. It relates the Bassam quantum time to its quantum information entropy (von Neumann entropy), which is a dimensionless quantity. The theory also presents Equation (53), termed Bassam's fundamental spatial equation, which relates the Bassam quantum length to its quantum information entropy (von Neumann entropy), which is a dimensionless quantity. These

equations are consistent with several fundamental principles in modern theoretical physics.

This theory proposes that time is not a fundamental entity but rather emerges and arises from quantum information [52]. Here, S_{VN}, which represents the quantum information entropy (von Neumann entropy), is linearly related to time, indicating that the perceived flow of time is an evolution of the information content of the system [53].

Upon applying Equation (47) to the photon in a pure polarization state, $S_{VN}^{(Ph)} = 0$, we obtain $t_B^{(Ph)} = 0$, where the theory predicts an explanation of time freezing at the speed of light in special relativity. This leads to the vanishing of the information entropy and to the vanishing of internal time. Upon applying Equation (47) to the photon, we obtain $L_B^{(Ph)} = 0$, which implies the vanishing of the internal spatial extension of the photon. This explains that the photon traveling at the speed of light is not merely massless; rather, it is due to a deeper reason than that. And it is that the photon does not possess quantum information entropy, which means that the photon does not experience a time of its own nor step in space. Therefore, its speed is considered the maximum speed, in agreement with the absence of a preferred rest frame for the photon [54].

4. 2 COMPARISON WITH CURRENT MODELS OF TIME AND SPACE

This theory differs from traditional models of time in that it is characterized by time emerging through S_{VN}, which represents the quantum information entropy (von Neumann entropy), π is a mathematical constant. In comparison with Einstein's general relativity, where time emerges through the curvature of spacetime, t_B in this theory is an intrinsic feature of the quantum state of the system itself, suggesting that time pre-exists geometry [55]. The emergence of the Bassam length $L_B^{(e)}$ and Bassam time $t_B^{(e)}$ as a fundamental quantum scale for the electron points to spacetime pixels or a fine lattice network, offering a solution to the problem of singularities in general relativity. The existence of a minimum limit for the Bassam quantum length L_B prevents spacetime from collapsing to a point of infinity. Moreover, the Bassam coordinates make spacetime a discrete network instead of a continuous fabric. where physics moves from models of describing phenomena to models of explaining the origin of these phenomena that worked to form the laws of physics. Thus, within the framework of this theory, the universe is a network of fundamental indivisible pixels. The quantized Bassam coordinates can be written (upon restoring SI units), which are given by the following relation.

$$(t_B\,, x_B\,, y_B, z_B) =$$

$$\left(n_t \frac{S_{VN}}{\pi}\frac{\hbar}{mc^2}, n_x \frac{S_{VN}}{\pi}\frac{\hbar}{mc}, n_y \frac{S_{VN}}{\pi}\frac{\hbar}{mc}, n_z \frac{S_{VN}}{\pi}\frac{\hbar}{mc}\right) \quad (81)$$

where S_{VN}, which represents the quantum information entropy (von Neumann entropy), is a dimensionless quantity. By setting ($1 = n_t = n_x = n_y = n_z$), we obtain the Bassam coordinates that represent fundamental indivisible pixels in the spacetime network, given by the following relation.

$$(t_B\,, x_B\,, y_B, z_B) =$$

$$\left(\frac{S_{VN}}{\pi}\frac{\hbar}{mc^2}, \frac{S_{VN}}{\pi}\frac{\hbar}{mc}, \frac{S_{VN}}{\pi}\frac{\hbar}{mc}, \frac{S_{VN}}{\pi}\frac{\hbar}{mc}\right) \quad (82)$$

We observe that in this theory the term $2\pi^2/\ln 2$, in Equation (69) is half of the constant $4\pi^2/\ln 2$ derived by Hernandez et al. [56] in the context of photonic black holes. Now we can obtain a fundamental coefficient that indicates the common origin of information in particle physics and gravitational physics, and it can be termed the Bassam coefficient for information, which is given by the following equation.

$$\mathcal{B}_{Bassam} = 1 + \frac{2\pi^2}{S_{VN}} n \quad (83)$$

where $\mathcal{B}_{Bassam}$ is the Bassam coefficient for information, and it is not a fixed numerical constant. S_{VN} is the quantum information entropy (von Neumann entropy), and it is a dimensionless quantity. For the electron,

where $S_{VN} = \ln 2$, we obtain the following equation.

$$\mathcal{B}^{(e)}_{Bassam} = 1 + \frac{2\pi^2}{\ln 2} = 29.478 \qquad (84)$$

However, in thermodynamics, the thermodynamic arrow of time is associated with the increase of thermodynamic entropy in an open system. As for t_B, it is associated with the quantum information entropy (von Neumann entropy), which distinguishes the time t_B from the statistical thermodynamic time [57].

In the framework of classical general relativity, gravitational collapse culminates in a spacetime singularity of infinite density. However, within Bassam's Unified Quantum Theory, continuous spacetime is replaced by a discrete, quantized network. The existence of the fundamental minimum length scale—the Bassam quantum length (L_B)—inherently prevents this infinite collapse. Consequently, the classical singularity is replaced by a finite, quantized entity at the Bassam length scale, which we term the Bassam Quantum Core. This theoretical core resolves the singularity problem, suggesting that the black hole interior is governed by quantum information entropy rather than an infinite void.

4. 3 TESTABLE PREDICTIONS

First, this theory provides a new time scale at the quantum level for the electron, given by the value $t_B^{(e)} \approx 2.846 \times 10^{-22}$s. which indicates that $t_B^{(e)}$ represents a new internal Bassam quantum time at the subatomic level [58]. It has not previously appeared in previous physical models. This indicates the existence of an internal temporal structure for particles, which opens prospects for understanding the nature of time in quantum mechanics.

Second, upon using Equation (53), the Bassam quantum length of the electron, $L_B^{(e)}$, can be obtained. It represents the smallest possible distance that an electron can traverse from one place to another as a quantum spatial extension, and it is given by the precisely calculated value $L_B^{(e)} \approx 8.520 \times 10^{-14}$m. This makes it a bridge between atomic physics and nuclear physics, and a natural bridge for understanding the spatial structure of particles at the subatomic level.

Third, upon using Equation (56), we can obtain a new atomic particle called the excited Bassam electron, which is given a mass of $m_B^{(e)} \approx 15.1\ \mathrm{MeV/c^2}$ (the precisely calculated value is $m_B^{(e)} \approx 15.063\ \mathrm{MeV/c^2}$.

Fourth, the theory predicts that a highly entangled quantum state with S_{VN}, which represents a large entropy, possesses a faster internal time scale, in in which t_B is small; this can be tested through quantum interference experiments using entangled photon pairs [59].

5. CONCLUSION

This theory has proposed that time for any physical system is not an absolute entity; rather, time emerges through its derivation from the quantum information content via Equation (47), which was applied to the electron in the maximally mixed state $S_{VN}^{(e)} = \ln 2$, thereby yielding the existence of the internal Bassam quantum time, given by the value $t_B^{(e)} \approx 2.846 \times 10^{-22}$s. An internal quantum explanation for the freezing of time at the speed of light is also provided by the theory, leading to the vanishing of the quantum information entropy of the photon, $S_{VN}^{(Ph)} = 0$. Consequently, $t_B^{(Ph)} = 0$ is obtained, which represents the internal Bassam quantum time of the photon. This theory has explained that the photon travels at the speed of light not only because it is massless in the traditional interpretation; rather, it has provided a deeper explanation than that, in that the photon travels at the speed of light because it does not possess quantum information entropy, and therefore it travels at the maximum speed. This theory also presents Equation (53) for space, and Equation (68), the mass spectral equation, which leads to the emergence of new particles. This theory is distinguished by opening new horizons in modern theoretical physics.

5. 1 IMPORTANCE OF THE THEORY

This theory provides the first connection between quantum mechanics, relativity, and

quantum information theory in modern theoretical physics, contributing to solving the problem of time [61]. It does not limit itself to providing a law for time t_B, but rather presents a parallel law for space L_B. Experimentally, it can provide quantitative predictions that are testable in quantum systems and entangled states [61]. This theory is distinguished by describing the nature of the relationship between time and information as two sides of a single physical coin, connected with quantum reality, and this theory links information, geometry, and mass in modern theoretical physics [61].

REFERENCES

[1] J. Barbour, The Nature of Time, arXiv: 0903.3489 [gr-qc] (2009).
[2] C. Marletto and V. Vedral, "Evolution without evolution, and without ambiguities," Phys. Rev. D 95, 043510 (2017).
[3] L. Nye, "The Emergence of Time from Quantum Information Dynamics," J. High Energy Phys. Grav. Cosmol. 10, 1981 (2024).
[4] E. Schrödinger, "Quantization as an Eigenvalue Problem" [Quantisierung als Eigenwertproblem], Ann. Phys. (Leipzig) 384, 361 (1926).
[5] P. A. M. Dirac, "The quantum theory of the electron," Proc. R. Soc. Lond. A 117, 610 (1928).
[6] G. Chiribella and C. M. Scandolo, "Conservation of information and the foundations of quantum mechanics," EPJ Web of Conferences 95, 03003 (2015).
[7] E. Noether, "Invariante Variationsprobleme," Nachr. Ges. Wiss. Göttingen, Math.-Phys. Kl., 235 (1918).
[8] L. D. Landau and E. M. Lifshitz, *The Classical Theory of Fields* (Butterworth-Heinemann, Oxford, 1975).
[9] L. Susskind, "The world as a hologram," J. Math. Phys. 36, 6377 (1995).
[10] D. Harlow, "Jerusalem lectures on black holes and quantum information," Rev. Mod. Phys. 88, 015002 (2016).
[11] A. Einstein, "The Foundation of the General Theory of Relativity," Ann. Phys. 49, 769 (1916).
[12] J. van Dongen, *Einstein's Unification* (Cambridge University Press, Cambridge, 2010).
[13] C. W. Misner, K. S. Thorne, and J. A. Wheeler, *Gravitation* (W. H. Freeman, San Francisco, 1973).
[14] A. Einstein, "Zur Elektrodynamik bewegter Körper," Ann. Phys. 17, 891 (1905).
[15] W. Rindler, *Introduction to Special Relativity* (Oxford, 1991).
[16] W. Rindler, "Hyperbolic motion in curved space-time," Phys. Rev. 119, 2082 (1960).
[17] M. Parikh, "New coordinates for de Sitter space," Phys. Lett. B 546, 189 (2002).
[18] R. Bousso, "Holography in general space-times," JHEP 06, 028 (1999).
[19] J. D. Bekenstein, "Black holes and the second law," Nuovo Cim. Lett. 4, 737 (1972).
[20] S. W. Hawking, "Gravitational radiation from colliding black holes," Phys. Rev. Lett. 26, 1344 (1971).
[21] J. D. Bekenstein, "Generalized second law," Phys. Rev. D 9, 3292 (1974).
[22] R. M. Wald, *General Relativity* (University of Chicago Press, Chicago, 1984).
[23] J. D. Bekenstein, "Black holes and entropy," Phys. Rev. D 7, 2333 (1973).
[24] K. Schwarzschild, "Über das Gravitationsfeld eines Massenpunktes nach der Einsteinschen Theorie," Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften, 189 (1916) [English translation: Gen. Relativ. Gravit. 35, 951 (2003)].
[25] M. M. Vopson, "The mass–energy–information equivalence principle," AIP Advances 9, 095206 (2019).
[26] C. Cohen-Tannoudji, B. Diu, and F. Laloë, *Quantum Mechanics, Vol. 1* (Wiley, New York, 1977).
[27] J. D. Jackson, *Classical Electrodynamics*, 3rd ed. (Wiley, New York, 1999).
[28] D. J. Griffiths, *Introduction to Electrodynamics*, 4th ed. (Cambridge University Press, Cambridge, 2017).
[29] C. Cohen-Tannoudji, J. Dupont-Roc, and G. Grynberg, *Atom-Photon Interactions: Basic Processes and Applications* (Wiley, New York, 1992).

[30] R. P. Feynman, *QED: The Strange Theory of Light and Matter* (Princeton University Press, Princeton, 1985).
[31] A. Einstein, "Does the inertia of a body depend upon its energy content?," Ann. Phys. 18, 639 (1905).
[32] B. K. Bagchi, *Partial Differential Equations for Mathematical Physicists* (CRC Press, Boca Raton, 2020).
[33] R. J. Nagem, "Wave propagation with local conservation laws in discrete space-time," J. Acoust. Soc. Am. 95, 2960 (1994).
[34] L. D. Landau and E. M. Lifshitz, *Quantum Mechanics* (Pergamon, 1977).
[35] S. W. Hawking, "Particle creation by black holes," Commun. Math. Phys. 43, 199 (1975).
[36] M. E. Peskin and D. V. Schroeder, *An Introduction to Quantum Field Theory* (Addison-Wesley, Reading, MA, 1995).
[37] T. Padmanabhan, "Gravity and the thermodynamics of horizons," Phys. Rept. 406, 49 (2005).
[38] R. Fitzpatrick, *Oscillations and Waves: An Introduction*, 2nd ed. (CRC Press, Boca Raton, 2018).
[39] L. Lyons, *All You Wanted to Know about Mathematics but Were Afraid to Ask* (Cambridge University Press, Cambridge, 1995).
[40] D. J. Griffiths, *Introduction to Quantum Mechanics*, 3rd ed. (Cambridge University Press, Cambridge, 2018).
[41] A. Einstein, "Über einen die Erzeugung," Ann. Phys. 322, 132 (1905).
[42] F. Halzen and A. D. Martin, *Quarks and Leptons: An Introductory Course in Modern Particle Physics* (Wiley, New York, 1984).
[43] M. Srednicki, *Quantum Field Theory* (Cambridge University Press, Cambridge, 2007).
[44] F. Schwabl, *Advanced Quantum Mechanics*, 4th ed. (Springer, Berlin, 2008).
[45] T. Ohlsson, *Relativistic Quantum Physics: From Advanced Quantum Mechanics to Introductory Quantum Field Theory* (Cambridge University Press, Cambridge, 2011).
[46] J. von Neumann, *Mathematical Foundations of Quantum Mechanics* (Springer, 1932).
[47] E. Schrödinger, "Quantisierung als Eigenwertproblem (Zweite)," Ann. Phys. 384, 489 (1926).
[48] M. J. Duff, L. B. Okun, and G. Veneziano, "Trialogue on fundamental constants," JHEP 03, 023 (2002).
[49] A. Khrennikov and Y. Volovich, "Energy levels of 'hydrogen atom' in discrete time dynamics," Open Syst. Inf. Dyn. 13, 119 (2006).
[50] Belle II Collaboration, *The Belle II Physics Book*, PTEP 2019, 123C01 (2019).
[51] J. P. Lees et al. (BaBar Collaboration), "Search for a Dark Leptophilic Scalar in e^+e^- Collisions," Phys. Rev. Lett. 125, 181801 (2020).
[52] C. Rovelli, "Relational quantum mechanics," Int. J. Theor. Phys. 35, 1637 (1996).
[53] D. N. Page and W. K. Wootters, "Evolution without evolution: Dynamics described by stationary observables," Phys. Rev. D 27, 2885 (1983).
[54] W. G. Unruh and R. M. Wald, "Information Loss," Rep. Prog. Phys. 80, 092002 (2017).
[55] Y. Nomura, N. Salzetta, F. Sanches, and S. J. Weinberg, "Spacetime equals entanglement," Phys. Lett. B 763, 370 (2016).
[56] X. Hernandez, C. S. Lopez-Monsalvo, S. Mendoza, and R. A. Sussman, "Some statistical mechanical properties of photon black holes," *Rev. Mex. Fis.*, vol. 52, no. 6, pp. 515, 2006.
[57] R. Balian, "Information in statistical physics," Stud. Hist. Phil. Sci. Part B 36, 323 (2005).
[58] E. Recami, "Elementary time and the physical meaning of the Compton time," Found. Phys. 13, 341 (1983).
[59] A. M. Kaufman, M. E. Tai, A. Lukin, M. Rispoli, R. Schittko, P. M. Preiss, and M. Greiner, "Quantum thermalization through entanglement in an isolated many-body system," Science 353, 794 (2016).
[60] M. S. Klinger and R. G. Leigh, "The Problem of Time and its Quantum Resolution," arXiv:2504.00152 (2025).
[61] M. A. Salih, "Gravitational interactions with information dynamics," Front. Astron. Space Sci. 12, 1647284 (2025).